PS 1

$E(\ ,\ )$ ENERGY FUNCTIONAL

x_1 – LENGTH OF STIFF SPRING

$\bar{x}_1$ – REST LENGTH OF STIFF SPRING

$\Delta x_1 = \bar{x}_1 - x_1$

ALTERNATIVELY:



x_2 – POSITION OF END OF SOFT SPRING

22) $x_1 = \bar{x}_1 - \frac{F}{k_1}$

$x_2 = \bar{x}_2 - \frac{F}{k_2} = \bar{x}_2 - \frac{F}{\epsilon k_2}$ ← MORE BY ϵ

iii) SAME TENSION

iv) ENERGIES

$E_1 = \frac{1}{\epsilon} E_2$

? SPRING 2 STORES MORE

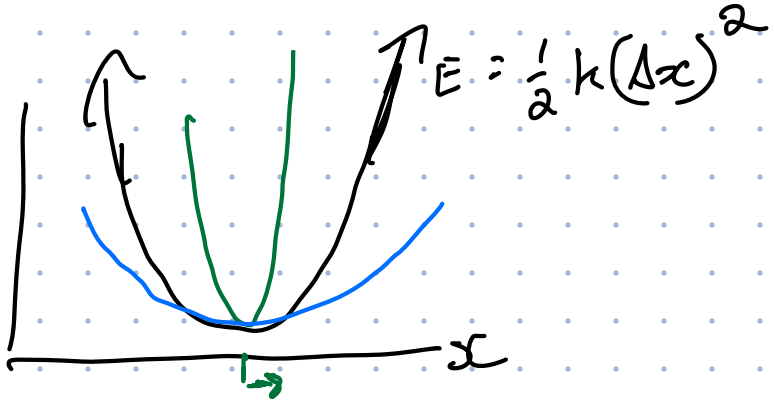
?

$E_1 = \epsilon E_2$

$\epsilon E_1 = E_2$

$E = \frac{1}{2} k (\Delta x)^2$

IDEALIZATION



REAL?

